with corresponding eigenvectors

$$v,\ \overline{v} \quad (Av = \alpha v,\ A\overline{v} = \overline{\alpha v}).$$

Choose a basis

$$\{e_1 = v + \overline{v},\ e_2 = \sqrt{\varepsilon}(v - \overline{v})\}.$$

In this basis, the matrix A will have the form

$$\begin{pmatrix} \alpha_1 & \varepsilon\alpha_2 \\ \alpha_2 & \alpha_1 \end{pmatrix},$$

justifying the description of representative elements of this conjugacy class.

In the basis $\{v, \overline{v}\}$, matrices that commute with A will have the form

$$\begin{pmatrix} \lambda & 0 \\ 0 & \overline{\lambda} \end{pmatrix},$$

for all

$$\lambda \in \mathbb{F}_{q^2}^{\times},$$

so the number of such matrices is $q^2 - 1$.

5.25.2. 1-dimensional representations. First, we describe the 1-dimensional representations of G.

Proposition 5.25.1. $[G, G] = SL_2(\mathbb{F}_q)$.

Proof. Clearly,

$$\det(xyx^{-1}y^{-1}) = 1,$$

so

$$[G, G] \subseteq SL_2(\mathbb{F}_q).$$

To show the converse, it suffices to show that the matrices

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

are commutators (as such matrices generate $SL_2(\mathbb{F}_q)$). Clearly, by using transposition, it suffices to show that only the first two matrices are commutators. But it is easy to see that the matrix

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$